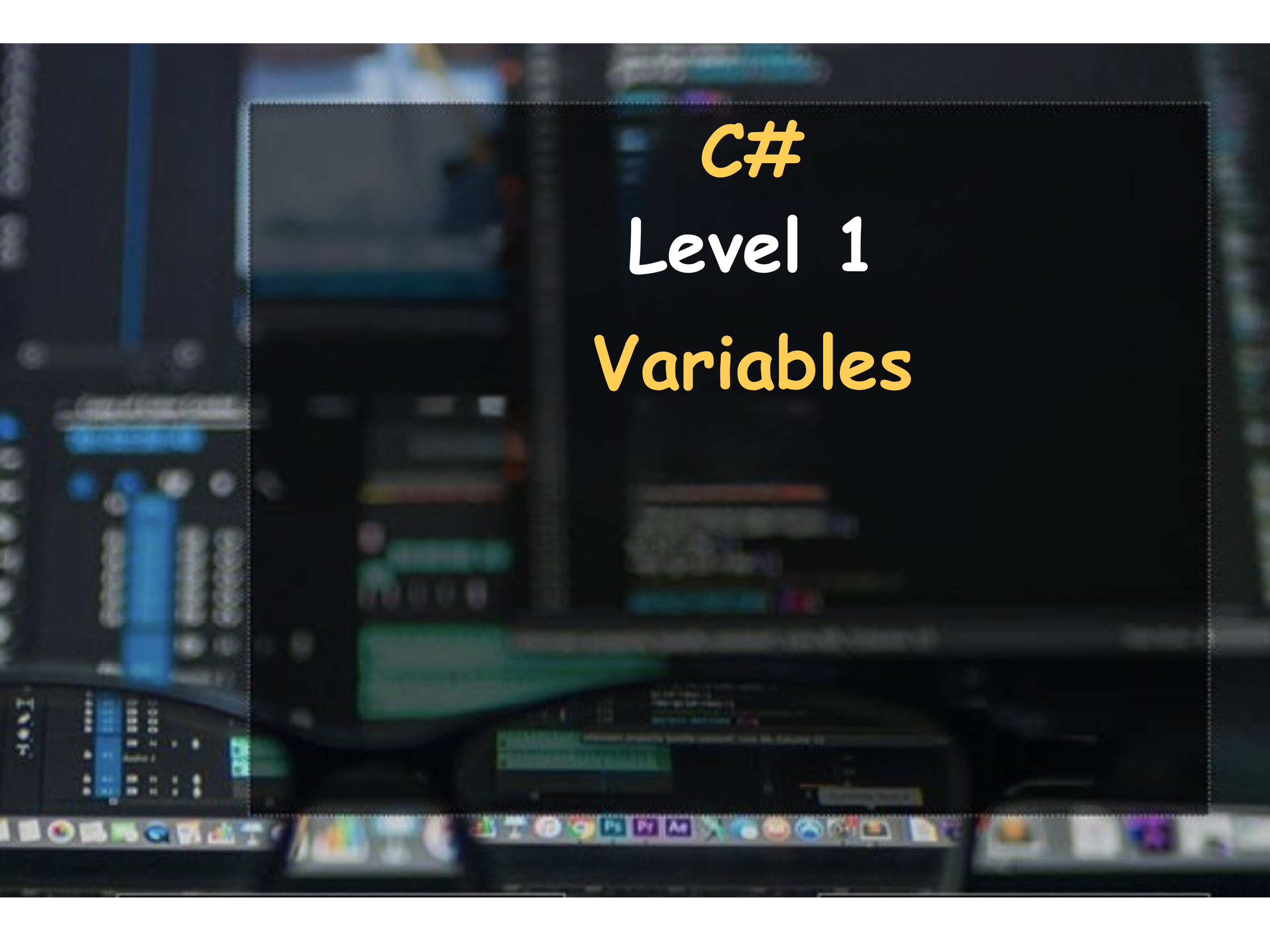




C#

Level 1

Variables



C#

Level 1

Variables

🔍 Definition

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- ◆ A variable is a named storage location used to hold a value in a program.
- ◆ Each variable has a data type, which determines what kind of value it can store.

Explanation

◆ The general syntax for declaring a variable is:

`DataType VariableName = Value;`

 `int` → Integer data type

 `x` → Variable

 `10` → Stored value

Procedures

1 Create a String Variable

```
string MyName = "Sara Alterkawi";
```

◆ A string stores text.

```
Console.WriteLine(MyName);
```

2 Create Integer Variables

```
int x = 10;
```

```
int y = 20;
```

◆ int is used to store whole numbers.

```
Console.WriteLine("x=" + x);
```

```
Console.WriteLine("y=" + y);
```

3 Adding Variables

◆ Be careful when using + with strings and numbers.

✗ This does not calculate the sum:

```
Console.WriteLine("x+y=" + x + y);
```

4 Calculate the Sum Correctly

✓ Use parentheses:

```
Console.WriteLine("x+y=" + (x + y));
```

◆ (x + y) is calculated first.

5 Create a double Variable

```
double MyDouble = 25.89D;
```

◆ double stores numbers that can contain decimal values.

Procedures

❑ Create a String Variable

❑ Create a char Variable

char MyLetter = 'M';

◆ char stores one character.

✦ A char uses single quotes:
'M'

❑ Create a bool Variable

bool MyBool = true;

◆ bool stores one of two logical values:

 true

 false

💡 Example

Common Data Types

Data Type	Stores	Example
string	Text	"Sara"
int	Whole numbers	10
double	Decimal numbers	25.89
char	One character	'M'
bool	True/False	true

⊘ Warning

⚠ Do not confuse addition with string concatenation.

`"x+y=" + x + y`

→ Produces:

`x+y=1020`

But:

`"x+y=" + (x + y)`

→ Produces:

`x+y=30`

⚡ The parentheses force C# to calculate `x + y` before joining the result with the string.



Comparison



int

- ✓ Whole numbers
- ✓ Example: 10, 25, -5



double

- ✓ Decimal numbers
- ✓ Example: 25.89, 3.14



char

- ✓ One character
- ✓ Uses ' '
- ✓ Example: 'M'



string

- ✓ Text
- ✓ Uses " "
- ✓ Example: "Sara"



bool

- ✓ Logical value
- ✓ true or false

✦✦ Summary

- ✓ Variables store values in a program
- ✓ Every variable has a data type
- ✓ string stores text
- ✓ int stores whole numbers
- ✓ double stores decimal numbers
- ✓ char stores one character
- ✓ bool stores true or false
- ✓ Use $(x + y)$ when you want arithmetic before string concatenation

Thank you 😊